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Saveetha Institute of Medical And Technical Sciences
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Hybrid Cryptography-Based Secure Messaging Application

CSA5111-CRYPTOGRAPHY AND NETWORK SECURITY

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Abstract

- The **Hybrid Cryptography-Based Secure Messaging Application** is designed to provide secure, confidential, and authenticated communication over public networks by combining both **asymmetric (RSA)** and **symmetric (AES)** cryptographic algorithms.
- The application uses RSA encryption for secure key exchange and AES-256 encryption for fast and efficient message encryption.
- User authentication is implemented using **PBKDF2/Bcrypt password hashing** and **JWT (JSON Web Token)** session management to ensure secure access control.
- Message integrity and authenticity are maintained through **HMAC-based digital signatures**, while encrypted chat history is securely stored to prevent unauthorized modification.

Modules Description

User Authentication

- Secure user registration and login using email/username and password.
- Passwords are hashed using **PBKDF2/Bcrypt** before storage.

. RSA Key Exchange

- Generates **2048-bit RSA public and private key pairs**.
- Securely exchanges the AES session key using RSA encryption.

AES-256 Message Encryption

- Encrypts chat messages using the **AES-256-GCM** symmetric encryption algorithm.
- Provides fast, efficient, and highly secure encryption for real-time messaging.

Message Integrity & Secure Chat History

- Verifies message authenticity using **HMAC-SHA256**.
- Detects any unauthorized modification or tampering of messages.



Engineer to Excel

Techniques and Tools Used



Techniques Used

RSA Encryption – Securely exchanges secret keys between users.

AES Encryption – Encrypts and decrypts chat messages quickly.

Password Hashing – Protects user passwords using PBKDF2/Bcrypt.

JWT Authentication – Provides secure user login and session management.

HMAC Verification – Checks that messages have not been changed.

Tools Used

Programming Language: Python

Frontend: HTML, CSS, JavaScript

Backend: Flask

IDE: Visual Studio Code / PyCharm

Database: SQLite / MongoDB

Libraries: cryptography, bcrypt, hashlib, PyJWT

Platform: Windows / Linux

Version Control: Git (Optional)



GPS Map Camera

Mevalurkuppam, Tamil Nadu, India 
22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026498° Long 80.017098°
Monday, 27/07/2026 11:51 AM GMT +05:30

